

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester VI Examination, 2013-14

Paper No. CHB-604: Physical Chemistry-IV

Time: 3 Hours

Maximum Marks: 70

Instructions: Answer five questions in total, including Question 1 which is compulsory.

Question 1. Answer all parts of the following: [5 × 2 = 10]

- (a) Protons of CH₃I appear at 130 Hz in a 50 MHz NMR instrument. At what frequency would these be observed in a 600 MHz instrument?
- (b) Explain the conditions a function should satisfy for becoming a well-behaved wave function in quantum chemistry.
- (c) Differentiate between 'state' and 'microstate' in statistical thermodynamics.
- (d) Write the expression for the vibrational partition function.
- (e) Differentiate between Fermi resonance and coupling in IR spectroscopy.

Question 2.

- (a) Write the Schrödinger wave equation for a one-dimensional harmonic oscillator. Write general expressions for its energy levels and wave functions. Sketch the lowest two wave functions. [8]
- (b) Write the time-dependent Schrödinger wave equation. Under which conditions can you transform it into the time-independent equation? [6]

Question 3.

- (a) What are the factors that determine the intensity of a spectral line? [4]
- (b) How would you classify rotating molecules on the basis of their three principal moments of inertia (I_A, I_B, I_C)? [4]
- (c) How would you evaluate the two bond lengths in the linear OCS molecule using microwave spectral data? [6]

Question 4.

- (a) Show that for an anharmonic oscillator, the selection rules are $\Delta v = \pm 1, \pm 2, \pm 3, \dots$. Why is only the transition originating from $v = 0$ considered? [6]

- (b) State the Franck-Condon principle and illustrate it using potential energy curves for the ground and excited electronic states. [8]
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Question 5.

- (a) What are the spin-lattice (T_1) and spin-spin (T_2) relaxation times in NMR spectroscopy? Explain their physical meanings. [6]
- (b) Define the relation $S = k_B \ln \Omega$. Using this, derive the translational molecular partition function of an ideal gas. [8]